NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-39035/2F
31 January 1994
SUPERSEDING
MIL-R-39035/2E
5 March 1987

PERFORMANCE SPECIFICATION

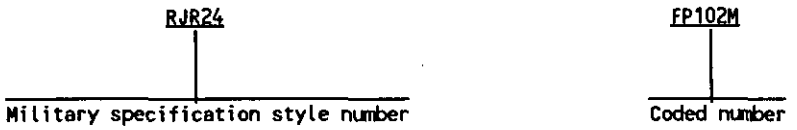
RESISTORS, VARIABLE, NONWIRE-WOUND (ADJUSTMENT TYPE, LEAD-SCREW ACTUATED), ESTABLISHED RELIABILITY STYLE RJR24

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RJR24, established reliability, adjustment type, lead-screw actuated, nonwire-wound, variable resistors. This style is available in characteristic C, F, and H.

1.2 Part or Identifying Number (PIN). Resistors covered by this specification shall be identified by a PIN which shall consist of a basis style of this specification and a coded number. The PIN shall be in the following form:



2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-R-39035 - Resistors, Variable, Nonwire-wound (Adjustment Type), Established Reliability, General Specification for.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELDM, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be in accordance with MIL-R-39035, and as specified herein.

3.2 Design and construction. The resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be 1/2 watt for all characteristics.

3.4 Terminals. Characteristics C, F, and H are available with L-, P-, W- and X- type terminals.

3.5 Nominal resistance value and maximum rated ac or dc working voltage. Nominal resistance values and maximum rated ac or dc working voltages shall be as specified in table I.

3.6 Actual effective electrical travel. Actual effective electrical travel shall be 15 turns minimum, and 30 turns maximum.

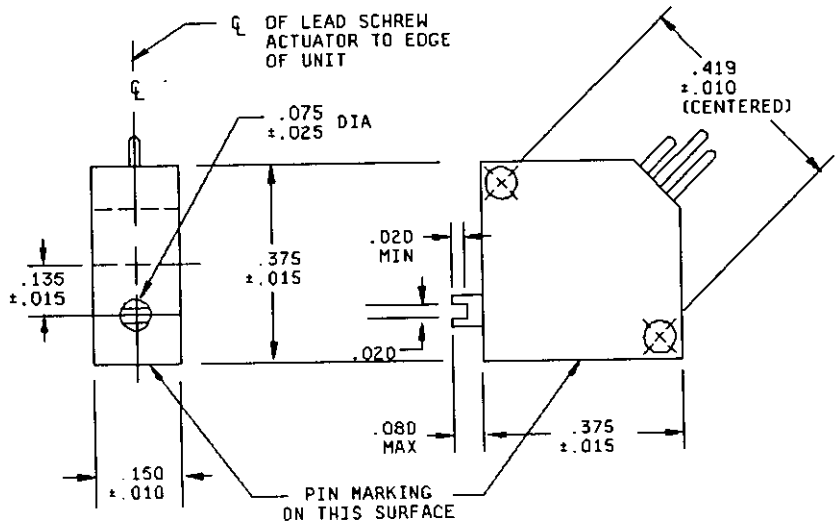
3.7 Operating torque. Operating torque shall be a maximum of 5 ounces-inches.

3.8 Maximum voltage. Maximum rated ac or dc working voltage shall be 300 volts.

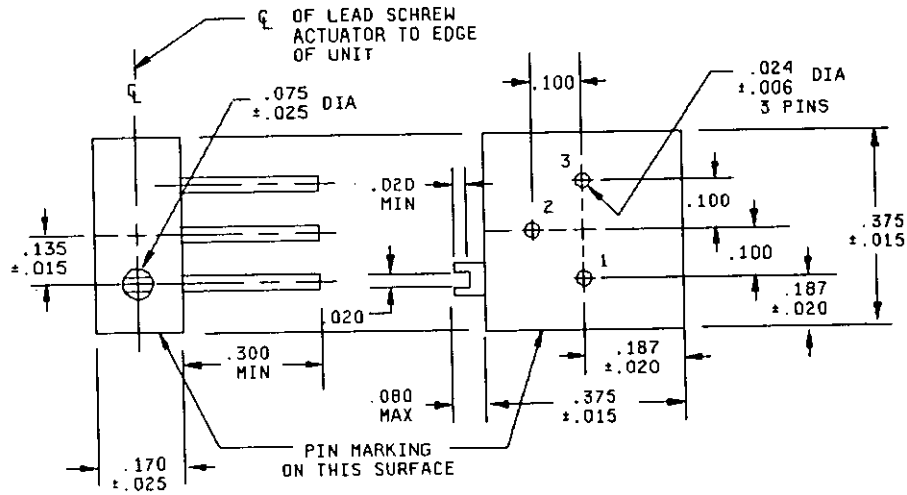
TABLE I. Nominal resistance value and maximum rated ac or dc working voltage.

Nominal resistance value	Maximum rated ac or dc working voltage per characteristics
	C, F, and H
Ohms	
10	2.23
20	3.1
50	5.0
100	7.0
200	10.0
500	15.8
1,000	22.3
2,000	31.6
5,000	50.0
10,000	70.7
20,000	100
25,000	111
50,000	158
Megohms	
0.10	223
0.20	300
0.25	300
0.50	300
1.00	300

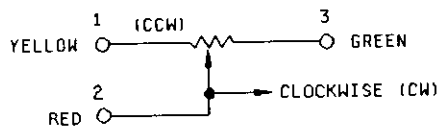
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TERMINAL TYPE L



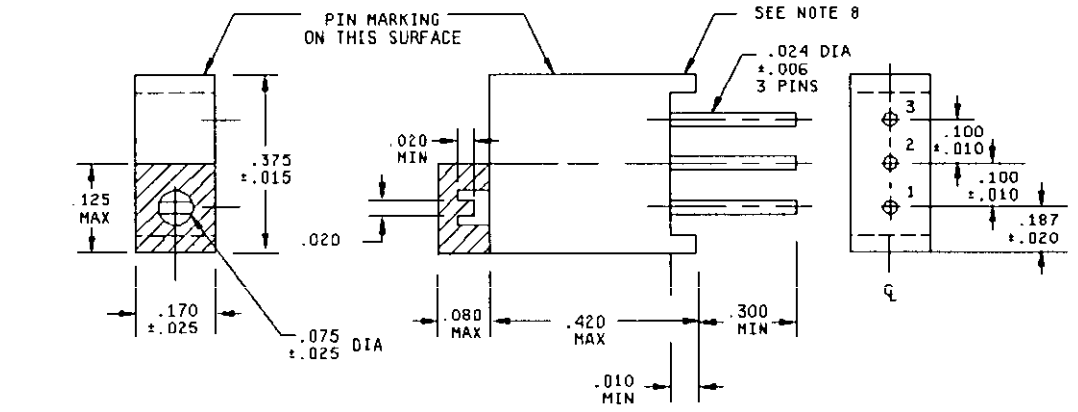
TERMINAL TYPE P



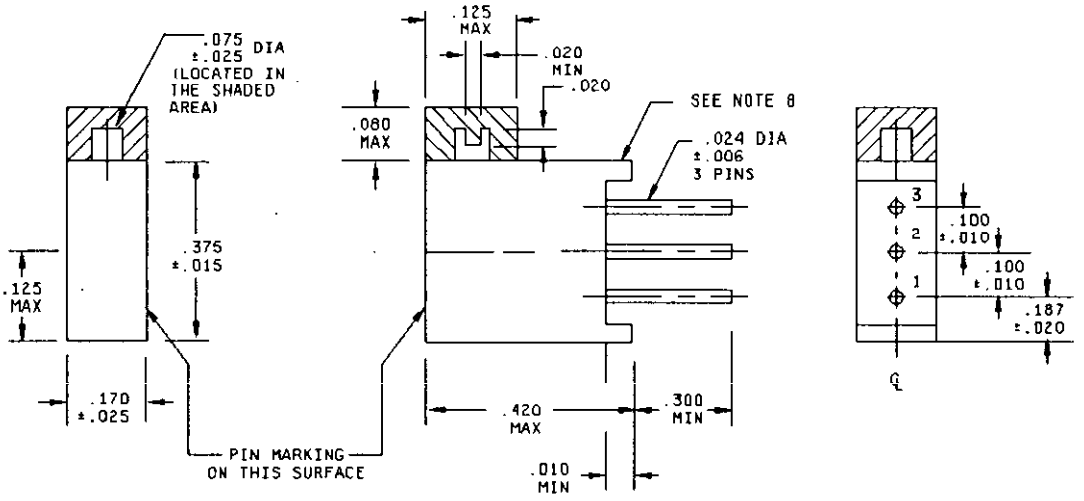
SCHMETIC DIAGRAM

FIGURE 1. Style RJR24 resistors.

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TERMINAL TYPE W



TERMINAL TYPE X

FIGURE 1. Style RJR24 resistors - Continued.

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Inches	mm	Inches	mm
.006	0.15	.105	2.67
.010	0.25	.125	3.18
.015	0.38	.135	3.43
.020	0.51	.150	3.81
.024	0.61	.170	4.32
.025	0.64	.187	4.75
.030	0.76	.300	7.62
.072	1.83	.375	9.53
.075	1.91	.419	10.64
.080	2.03	.420	10.67
.100	2.54		

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (.13 mm).
3. Metric equivalents are given for general information only.
4. The picturization of the styles above are given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope, and do not alter the functional aspects of the device are acceptable.
5. The entire slot of the actuating screw must be above the surface of the unit.
6. For types P, W, and X normal mounting means is by use of pins only.
7. The three leads shall be insulated wire, AWG size 28 to 30, having a minimum length of 6.00 (152.4 mm); they shall be insulated with polytetrafluoroethylene, stripped .250 ±.062 (6.35 ±1.57 mm) from the end and color coded.

FIGURE 1. Style RJR12 resistors - Continued.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-39035, and as specified herein.

4.2 Dielectric withstanding voltage. The magnitude of test voltage shall be 900 volts for all characteristics at atmospheric pressure, and 350 volts for all characteristics at reduced barometric pressure.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-R-39035 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

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6.3 Weight. The weight is .046 ounces (1.3 grams).

6.4 MIL-R-22097 substitution data. Resistors of this specification regardless of their failure rate designation are substitutes for resistors of the same resistance value, tolerance, terminal characteristic and resistance temperature characteristics specified in the inactivated specifications MIL-R-22097/3 and MIL-R-22097/4.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, AT, AV, ME, MI
Navy - AS, EC, OS, MC
Air Force - 17, 19, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-39035/2F	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
RESISTORS, VARIABLE, NOWIRE-WOUND (ADJUSTMENT TYPE, LEAD-SCREW ACTUATED), ESTABLISHED RELIABILITY STYLE RJR24

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME 645 LOG/LGMEP	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6696	(2) AUTOVON 986-6696
c. ADDRESS (Include Zip Code) Gentile Station 1060 Hamilton Street Dayton, OH 45444-5400	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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